

Q2: Countries in Sub-Saharan Africa will never overtake Luxembourg in GDP per capita terms. Discuss.

Luxembourg is a well-documented example of a high-income nation, with recent estimates of per-capita GDP hovering around \$129,000 in current US dollars with a population just above 660,000 (World Bank, n.d.). In contrast, Sub-Saharan Africa (SSA)¹ – home to nearly 2000 times Luxembourg’s population – records a per capita GDP measure of roughly \$1600 (World Bank, n.d.). At first glance, it appears to be highly improbable for SSA to overtake Luxembourg in GDP per capita terms. Yet, this essay draws on endogenous growth theory and models of technology transfer to outline conditions for convergence and to refute the claim that SSA nations will *never* overtake Luxembourg in GDP per capita terms. While obstacles exist, there is a plausible pathway for some nations to indeed converge and exceed Luxembourg’s income levels, provided certain policy conditions are met over a sufficient time horizon. The essay is organised as follows: first, theoretical mechanisms for convergence are discussed; next, evidence on technology diffusion, trade and market openness is introduced; then, structural obstacles faced by SSA are outlined; finally, recent initiatives and policy recommendations are assessed to conclude that prospects of catch-up do indicate that the gap can be closed, albeit challenging.

The proposition that Sub-Saharan African nations will *never* overtake Luxembourg in GDP per capita terms relies on several key assumptions. Firstly, the term “never” implies an absolute 0% probability of convergence, suggesting that even endogenous models will not allow for SSA economies to catch up and surpass Luxembourg’s income levels. Effectively, then, differences in GDP per capita terms must arise solely from exogenous and static variables.

¹ For the purpose of this essay, we define ‘Sub-Saharan Africa’ in line with the World Bank’s definition as provided here: <https://datahelpdesk.worldbank.org/knowledgebase/articles/906519-world-bank-country-and-lending-groups>

A simplistic view of such could be characterised by the Solow model, which explains cross-country differences in income levels by variance in savings rates and exogenously defining technology (Solow, 1956). However, much literature has since covered the shortcomings of the Solow model in accurately predicting observed income levels, partly due to the exogeneity of technology *A* (Mankiw et al., 1992).

To address these shortcomings, endogenous growth theory models are introduced. Romer (1990) presents a theoretical framework in which economic growth is driven by endogenous technological processes of ideas, innovation and knowledge. Within this framework, technology is *created* given research and development (R&D) and can exhibit increasing returns to scale. Technology is *nonrival*, since once produced, a design can be applied for infinitely many uses, yet it is simultaneously *excludable* under intellectual property laws. Within these confines, nations with high knowledge stock enjoy self-sustaining output growth through investment in human capital and R&D, which weakens the case for the possibility of convergence. However, the Romer model ignores the possibility of trade and technology transfer between rich nations, such as Luxembourg, and poorer nations, such as those of SSA. Without the inclusion of technology transfer mechanisms, a lagging region might *never* catch up if it remains trapped in a cycle of low innovation.

Howitt (2000) identifies forces that could support convergence through the proliferation of cross-country knowledge spillover. When ideas and technologies flow from a leading country to followers ('latecomers'), the poorer nations have the *advantage of backwardness*. Schumpeterian models predict that a nation far from the cutting edge of technology, or the technology frontier, can grow faster by imitating and catching up to technological advancements on a sector-by-sector basis. When a backward sector adopts frontier technology, it achieves a larger jump in productivity than a frontier country would, enabling higher growth

rates. The mechanism of creative destruction, thus, can imply conditional convergence if SSA nations are able to ‘absorb’ advanced technologies and outpace Luxembourg’s growth rate. In this manner, the convergence is conditional upon the ability of SSA nations to effectively absorb knowledge spillovers and take advantage of them. Absorptive capacity, hence, becomes the deciding variable in such cases of catch-up, dependent on the technological and managerial ‘know-how’ (such as the ‘tacit knowledge’ effect commonly attributed to Polanyi (1967)) as well as existing infrastructure levels of an economy (Cohen & Levinthal, 1989). Thus, theory suggests that SSA economies could grow faster than a mature economy such as Luxembourg if and only if they develop the capacity to absorb and implement the knowledge, failing which increasing returns to scale on innovation and human capital will favour the frontier country, entrenching divergence.

Having outlined theoretical models of convergence, the essay now analyses channels of catch-up. Channels of catch-up refer to mechanisms based on which poorer nations can move closer to the technological frontier. One critical channel of catch-up is international technology transfer, referring to the potential for SSA nations to acquire cutting-edge technology through relatively little investment in R&D themselves, supported by trade and foreign direct investment (FDI). Since technology is ever-evolving, there is a constant incentive for richer (North) nations to innovate in order to outcompete poorer (South) nations from imitating existing technology (Coe et al., 1997). Developing countries greatly benefit from the R&D performed in advanced countries, in turn, by being able to import capital goods through open trade and “borrow” the productivity gains made by new technologies (Keller, 2004). The primary conduits of technology diffusion are thought to be FDI and bilateral trade, often more important than domestic R&D for latecomer nations. Recent panel data analysis in Asian economies demonstrates a high degree of causal relationship between Foreign Direct Investment (FDI) on R&D measures based on established financial capital and technological

and managerial expertise (Erdal & Göçer, 2015). Beyond influencing initial knowledge stock, FDI inflows could reduce the heavy initial costs associated with establishing effective R&D protocols for SSA nations. As for international trade, similar endogenous growth models output a net world growth rate of 3% in a model without restrictions on trade flows between party nations (Rivera-Batiz & Romer, 1991, pp. 985). These findings suggest that should SSA economies deeply integrate with the global innovation system, they could accelerate their technological progress substantially, allowing for the possibility of convergence with frontier nations such as Luxembourg (Coe & Helpman, 1995).

The exact scale of the latecomer advantage from technology transfer depends on the local rate of absorption as mentioned earlier. Empirical research often produces mixed results such as the previously cited work by Erdal & Göçer (2015), who note that East Asian nations successfully leveraged foreign technology to enable convergence. On the other hand, Aghion & Howitt (1992) outline a mechanism through which, without innovation effort, simply importing technology might not be able to sustain growth since there is an asymptotic limit on productivity gains through 'imitative stagnation'. The flow of technology into poorer nations is heavily contingent upon the attractiveness of the economies for trade or FDI. So, in practice, SSA nations may not be able to substantiate the innovation levels required to ensure sustained and permanent growth in productivity levels. Grossman & Helpman (1991) formalise the notion that openness in foreign trade minimise externalities of scientific and technological knowledge that otherwise hurt economic growth and welfare of a model economy. Sachs & Warner (1997) explicitly note the lack of openness in many African economies as a key reason for the lacklustre growth rates of the region. After accounting for controls such as geography and initial levels of income, the following criteria were associated with higher growth: low tariffs, few quotas and export orientation. Openness to international trade, thus, is a necessary condition for convergence.

Trade integration has the additional benefit of facilitating indirect knowledge flows, implying that reducing trade frictions as in Sachs & Warner (1997) carries the advantage of speeding up the *velocity* of flow of ideas (Eaton & Kortum, 2002). It is necessary to note, however, that there may be reverse causality associated with the relationship between openness and growth (richer nations might be more prone to trade). For SSA economies to effectively leverage technology spillovers, institutional precursors are key. Easterly et al. (2004) note the weak correlations between aid inflows and growth and, while not directly comparable, emphasise the necessity of further considerations when assessing the validity blanket claims. Since many SSA nations are primarily exporters of commodities, they remain susceptible to exogenous price shocks unless they build resilient manufacturing or service sectors (World Bank, n.d.). The key is that openness provides an opportunity that is necessary but insufficient for SSA economies to catch-up to and surpass the income levels of Luxembourg. It must be paired with domestic reform and absorptive capacity-building policies to translate into *permanent* and *sustained* growth. Nevertheless, the theoretical models and empirical works discussed solve for a pathway in which global integration could foster higher growth by enabling competition and idea flows.

In practical terms, Luxembourg's high GDP per capita can be explained by the innovation-friendly and open market it enjoys, evidenced by its high ranking in the Global Innovation Index (eds. World Intellectual Property Organization et al., 2024). Further, Luxembourg's relatively open markets and high rankings in intellectual property rights index align with the primary takeaways outlined by Grossman & Helpman (1991) (ed. World Intellectual Property Organization, 2024). Investment in R&D is a large differentiator, with Luxembourg investing 1.04% of GDP in 2021, contrasted with SSA nations who invest between 0.3-0.5% of GDP in R&D. Even within SSA, there exists a wide variety of investment figures, ranging from 0.56% of GDP for Botswana, to 0.01% of GDP for Mauritania (World

Bank, n.d.). The apparent domestic knowledge creation deficit means that if SSA cannot reasonably improve its innovation capacity, endogenous growth theory would imply that overtaking a technology frontier nation such as Luxembourg is implausible. Factoring in trade openness and FDI, Luxembourg notes an export-to-GDP ratio of 212%, the highest in the world, stemming from its integration in the global financial services industry – a high-value-added activity. Conversely, the SSA average hovers around 25% for the same metric (World Bank, n.d.). Human capital and infrastructure paint a similar story. While Luxembourg enjoys near 100% literacy, universal electricity access and advanced infrastructure, only about 51% of SSA's population has access to electricity. Expected years of schooling in Luxembourg far outpace the mean in SSA with minimal tertiary enrolment (United Nations DP, n.d.). These gaps should serve analogous to absorptive capacity limitations experienced by the vast majority of SSA nations. Empirical standings show how Luxembourg maintains itself at the technological frontier, and how SSA lags far behind. However, this does not indicate that the gap cannot be closed, only that the scale of the challenge is wide.

Structural and institutional barriers that exacerbate the scale of convergence typically stem from weak governance that is exogenous, *yet static*. The implication is that while endogenous growth theory does not account for institutions that discourage investment, pervasive corruption, political instability or extractive regimes, these institutions are dynamic and subject to change. While Sachs & Warner (1997) find statistically significant geographic effects on growth rates, even after controlling for policies, the seminal paper by Acemoglu et al. (2001) argues that such institutions are the fundamental cause of long-run differences and outweigh the geographical determinism proposed by many before including Diamond (1997). That said, geography, too, can be overcome with technology provided resources are put to use efficiently with good governance (Jedwab & Moradi, 2016). Recent developments do indicate significant upside potential for growth in SSA. The African Continental Free Trade Area

(AfCFTA) is expected to generate a 62% increase in high-value-added exports a near doubling of intracontinental exports (World Bank, n.d.). Digital infrastructure innovation is on the rise, and while there is little high-quality research measuring its impact, estimates note a statistically significant improvement in markers of human capital, and in turn, absorptive capacity (Tiony & Yin, 2023).

To summarise, given sufficient institutional reforms are effectuated to maximise channels of technology diffusion, trade and market openness, endogenous growth theories with trade and technology transfer models allow for countries in Sub-Saharan Africa to converge and overtake Luxembourg in terms of GDP per capita.

(Word Count: 1997 including footnote)

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